

# Keruing Group: Izin Consent and Preference Service

## Detailed Design

*Architecture, requirements, and validation criteria for build*

|                     |                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Version             | 1.1                                                                                                                                                        |
| Status              | Design phase, build not started                                                                                                                            |
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| Companion documents | Izin Conceptual Design v0.8; Group Purpose Register GPR-3; Data Protection Impact Assessment DPIA-2026-11 (draft); Partner Programme Commercial Terms 2026 |

### Revision history

| Version | Date       | Changes                                                                                                                                                                                                                                                                                                                              |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 2026-09-24 | First consolidated detailed design.                                                                                                                                                                                                                                                                                                  |
| 1.1     | 2026-10-02 | Post-review revisions: FR-9; topic key (14.2, DEC-02); DNC result validity and refresh cycle (16, DEC-06); receipt links and receipt page (17); Eligibility API capacity (24.2, NFR-8); receipt storage (24.3); migration of customers with no recorded preference (25, DEC-11); acceptance criteria for FR-9, NFR-2 and NFR-8 (28). |

### Changes since version 1.0

This version updates the sections listed below after the design review of 2026-09-29. Sections not listed are unchanged from version 1.0.

- FR-9 and Section 17: when receipts are sent, the receipt link and what the receipt page shows.
- Section 14.2 and DEC-02: the key of the consent .changes topic.
- Section 16 and DEC-06: validity of DNC Registry results and the refresh cycle.
- Section 24.2 and NFR-8: campaign load on the Eligibility API and its provisioned rate.
- Section 24.3: storage for receipts.
- Section 25 and DEC-11: migration of customers with no recorded preference in the legacy systems.
- Section 28: acceptance criteria for FR-9, NFR-2 and NFR-8.

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## 1. Purpose and Scope

This document describes the architecture of Izin, the Keruing Group consent and preference service (CPS). Izin records what every Keruing customer has agreed to, for which purpose, through which channel and under which notice, and it tells every system that contacts customers or uses their data whether it may do so. It is written to a level of technical detail sufficient for an engineering team to begin implementation. Section 29 gives an explicit assessment of where that is and is not yet true.

Today each business unit keeps its own marketing flags: the telco CRM has three opt-out columns, the Rewards platform has one newsletter flag, and store sign-ups are recorded on paper forms that are scanned and keyed in by a vendor. A customer who withdraws in one place is still contacted from another, and the Office of the Data Protection Officer (ODPO) cannot show a regulator when or how any given consent was obtained. Izin replaces these flags with one purpose-based consent record per customer, one capture API for every channel, and one stream of consent changes that every downstream system applies.

### In scope

Consent capture for every Keruing channel (web, mobile app, contact centre, stores and partner channels); the purpose catalogue and notice versions; the consent record and its history; withdrawal; propagation of consent changes to downstream marketing, analytics, CRM and partner systems; the eligibility API that answers "may we contact this person for this purpose through this channel"; Do Not Call (DNC) Registry checks for Singapore telephone numbers; consent receipts; retention; the regulator evidence export; and migration of the legacy flags.

### Out of scope

The marketing campaign platform itself (Seruan, which consumes Izin); customer identity and authentication, which remain with Keruing ID; consent for cookies and web tracking, which the web team manages with its own banner and a separate design; and access and correction requests under the PDPA, which the ODPO handles through its case tool and which read from Izin through the regulator export in Section 20.

### Volume baseline

| Metric                                                  | 2026 actual | Design target (2028) |
|---------------------------------------------------------|-------------|----------------------|
| Customer profiles with a Keruing ID                     | 8.1 million | 8.7 million          |
| Active mobile lines (Keruing Mobile)                    | 5.6 million | 6.2 million          |
| Fibre broadband accounts (Keruing Fibre)                | 0.9 million | 1.1 million          |
| Keruing Rewards members                                 | 4.1 million | 4.8 million          |
| Consent changes per day (average)                       | 95,000      | 140,000              |
| Consent changes per second (peak, campaign reply surge) | 60          | 120                  |
| Marketing messages sent per month                       | 31 million  | 40 million           |

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## 2. Requirements

Every requirement below has a validation method and acceptance criterion in Section 28.

### 2.1 Functional Requirements

| ID    | Requirement                                                                                                                                                                                                    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1  | Capture consent and withdrawal per purpose from web, app, contact centre, stores and partner channels through one Consent API.                                                                                 |
| FR-2  | Record a consent as granted only from an affirmative act of the customer for that specific purpose; never from silence, inactivity or a pre-selected option.                                                   |
| FR-3  | Record for every change the channel, the actor (customer, contact centre agent, store staff, partner), the notice version shown, the time, and an evidence reference (session, call recording or signed form). |
| FR-4  | Offer withdrawal in every channel where consent can be given, with no more steps than granting.                                                                                                                |
| FR-5  | Publish every consent change to downstream consumers through the event bus.                                                                                                                                    |
| FR-6  | Provide an Eligibility API that answers whether a customer or contact point may be contacted for a purpose through a channel, combining consent state and DNC status.                                          |
| FR-7  | Check every Singapore telephone number against the DNC Registry before a marketing SMS or call is sent to it, unless the holder's clear and unambiguous consent is held in evidential form.                    |
| FR-8  | Provide a preference centre in web and app that shows every purpose, its current state and the notice text, and lets the customer change any of them.                                                          |
| FR-9  | Send the customer a consent receipt for every consent change, within five minutes of the change, by SMS to the mobile number on the profile or by email where the customer has no mobile number.               |
| FR-10 | Let approved partners capture consent for Keruing purposes in their own channels through the Partner Consent API, and supply partners with the consent state for the partner-sharing purposes.                 |
| FR-11 | Produce, on request of the ODPO, an evidence export for a customer, a purpose or a time range (Section 20).                                                                                                    |
| FR-12 | Treat an SMS reply of STOP to a Keruing marketing short code, and the opt-out option of the marketing IVR, as a withdrawal of the matching purpose.                                                            |
| FR-13 | Migrate the marketing preferences held in the legacy systems into Izin before the legacy flags are retired.                                                                                                    |
| FR-14 | Provide the ODPO with a console to manage purposes, notice versions and partner registrations.                                                                                                                 |

### 2.2 Non-Functional Requirements

| ID    | Requirement                                                                                                                                                                                                                                                    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1 | Consent API p99 latency of 300 ms or less for a single change at the design-target peak.                                                                                                                                                                       |
| NFR-2 | A withdrawal recorded in any channel is applied by every internal downstream consumer (Seruan, Group Analytics Platform, contact centre CRM and Rewards platform) within 24 hours, and is included in the next nightly partner file for the partner concerned. |
| NFR-3 | Availability of the Consent API and Eligibility API of 99.95% per calendar month.                                                                                                                                                                              |
| NFR-4 | No consent change acknowledged to a channel is lost on the failure of a node or an availability zone (RPO 0); objectives for the loss of the region are set in Section 23.2.                                                                                   |
| NFR-5 | For every consent in force, Izin can produce the evidence of how, when, through which channel and under which notice version it was given, with the full history of changes to it, within five business days of a request.                                     |
| NFR-6 | Personal data encrypted in transit and at rest; access on least privilege; every staff read of a customer record logged.                                                                                                                                       |
| NFR-7 | All Izin data stores reside in Singapore (AWS ap-southeast-1).                                                                                                                                                                                                 |
| NFR-8 | The Eligibility API sustains the design-target campaign peak of 1,000 requests per second with p99 latency of 50 ms or less.                                                                                                                                   |
| NFR-9 | No marketing SMS or call is made to a Singapore telephone number without either a valid DNC Registry result showing it is not registered, or clear and unambiguous consent held in evidential form.                                                            |

## 3. Foundational Principles

These principles govern every component in this document. Where a section has to trade them off, it says so.

1. **P1 Purpose-specific.** Consent is held per customer (or per contact point, Section 8) per purpose. There is no general "marketing" flag.
2. **P2 Affirmative act only.** A consent is recorded as granted only when the customer has taken a positive action for that purpose. Silence, inactivity, a pre-ticked box or acceptance of terms and conditions is never consent.

3. **P3 The event log is the system of record.** Every change is an immutable consent event. The current state is a projection of the events and can be rebuilt from them.
4. **P4 Withdrawal is never harder than granting.** A withdrawal takes effect in Izin at once and is answered by the Eligibility API on the next read.
5. **P5 Fail closed.** When consent or DNC status cannot be determined, the answer is "do not contact".
6. **P6 One truth.** Downstream systems keep no consent truth of their own; they apply Izin events or call the Eligibility API.
7. **P7 Minimum data.** Izin holds only the personal data needed to identify a customer and the contact points consents attach to.
8. **P8 Singapore resident.** Izin data stays in Singapore.

## 4. Business Units, Channels and Partners

| Business unit   | Customers                                             | Capture channels                                            | Marketing channels          |
|-----------------|-------------------------------------------------------|-------------------------------------------------------------|-----------------------------|
| Keruing Mobile  | 5.6 million active lines                              | App, web, contact centre, stores, SMS keyword               | SMS, voice, email, app push |
| Keruing Fibre   | 0.9 million accounts                                  | Web, contact centre, installer tablet (via the stores flow) | Email, voice                |
| Keruing Stores  | 61 stores, about 2.3 million identified buyers a year | Point-of-sale tablet, web                                   | Email, SMS                  |
| Keruing Rewards | 4.1 million members                                   | App, web, stores                                            | App push, email, SMS        |

Keruing Mobile allocates numbers from its own number ranges. A terminated mobile number is quarantined for six months and then returned to the pool for allocation to new subscribers.

Two partners take part in the 2027 partner programme:

- **Seri Assurance Berhad**, an insurer headquartered in Kuala Lumpur, offers device and travel insurance to Keruing customers who agree to purpose PTN-INS. Seri sells through its own app and its Kuala Lumpur contact centre.
- **Lintang Bank**, a Singapore bank, issues the Keruing Rewards co-brand credit card to customers who agree to purpose PTN-CARD.

The contact centre (about 900 agents across two sites in Singapore and one outsourced site) handles service calls and outbound telemarketing for all business units on one contact-centre platform with call recording.

## 5. Regulatory Basis

Izin is designed against the Personal Data Protection Act 2012 (PDPA) as amended in 2020, and the Commission's advisory guidelines. The provisions that shape the design are:

| Provision                     | What it requires                                                                                                                                                                                           | Where Izin addresses it |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Consent Obligation (s13, s14) | Consent for a purpose, given after notification; consent may not be made a condition of a product beyond what is reasonable                                                                                | Sections 7, 10          |
| Notification Obligation (s20) | Purposes notified on or before collection                                                                                                                                                                  | Section 7               |
| Withdrawal of consent (s16)   | Withdrawal on reasonable notice; the organisation informs the individual of the likely consequences and ceases use                                                                                         | Section 13              |
| DNC provisions (Part 9)       | Check with the DNC Registry before sending a specified message to a Singapore telephone number, unless clear and unambiguous consent in evidential form is held; give effect to withdrawal of that consent | Sections 13, 16         |
| Accountability (s11, s12)     | Policies, practices and the ability to demonstrate compliance                                                                                                                                              | Sections 19, 20         |



## Layer responsibilities

| Component           | Responsibility                                                                                                                                                                                     |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Consent API         | Accepts consent changes from first-party channels, validates them against the purpose catalogue, writes event, state and outbox rows in one transaction, returns the new state and receipt number. |
| Partner Consent API | Accepts consent changes from registered partners (Section 12) and forwards them to the Consent API with the partner as actor.                                                                      |
| Consent Store       | Aurora PostgreSQL 16 cluster holding events, state, contact points, purposes, notices, receipts and the outbox.                                                                                    |
| Outbox Relay        | Publishes outbox rows to the consent . changes topic (Section 14).                                                                                                                                 |
| Eligibility API     | Answers contact decisions from consent state and the DNC cache.                                                                                                                                    |
| DNC Service         | Keeps DNC Registry results for every Singapore number in the marketing base (Section 16).                                                                                                          |
| Receipt Service     | Issues consent receipts by SMS or email (Section 17).                                                                                                                                              |
| Partner Gateway     | Builds the nightly partner files from the event stream (Section 15).                                                                                                                               |
| ODPO Console        | Purpose and notice management, partner registry, evidence export (Section 20).                                                                                                                     |

Izin runs on Amazon EKS in the group's shared-services account, with the Consent Store on Aurora PostgreSQL and the event bus on Amazon MSK (Kafka).

## 7. Purpose Catalogue and Notice Versioning

The purpose catalogue is the single list of reasons for which Keruing collects, uses or discloses personal data that Izin governs. The ODPO owns it through the ODPO Console.

| Code       | Purpose                                                                                            | Basis            | Contact channel |
|------------|----------------------------------------------------------------------------------------------------|------------------|-----------------|
| MKT-SMS    | Marketing of Keruing products and offers by SMS                                                    | Consent          | SMS             |
| MKT-VOICE  | Marketing of Keruing products and offers by telephone call                                         | Consent          | Voice           |
| MKT-EMAIL  | Marketing of Keruing products and offers by email                                                  | Consent          | Email           |
| MKT-PUSH   | Marketing by app push notification and in-app message                                              | Consent          | Push            |
| PRS-OFFERS | Use of usage and purchase history to personalise offers across business units                      | Consent          | None            |
| PTN-INS    | Disclosure of name, mobile number, email and Rewards tier to Seri Assurance for insurance offers   | Consent          | Partner         |
| PTN-CARD   | Disclosure of name, mobile number, email and Rewards tier to Lintang Bank for co-brand card offers | Consent          | Partner         |
| SVC-NOTICE | Service and account messages                                                                       | Contract         | All             |
| LGL-REG    | Subscriber registration records kept under telecommunications regulation                           | Legal obligation | None            |

Each purpose has a series of notice versions. A notice version holds the exact customer-facing text in English, Chinese, Malay and Tamil, an effective date and a materiality flag set by the ODPO. A material change (a new category of data, a new recipient, a new channel) creates a new purpose version, and existing consents for the old version are not carried over: the customer is asked again, and until they answer, the purpose is treated as not granted: `consent_state` holds the purpose version of the latest event, and the Eligibility API answers a GRANTED state whose purpose version is older than the purpose's current version as not granted. A non-material change (wording, translation, layout) creates a new notice version under the same purpose version, and existing consents remain valid. Every consent event records the purpose version and notice version that the customer saw, so the evidence export can show the exact text.

Purpose codes are never reused. A retired purpose keeps its history and its events, but the Consent API rejects new grants for it.

## 8. Identity and Contact Points

A customer is identified by their Keruing ID subject (`mid`, a UUID issued by the group identity provider over OIDC). Online channels always carry the `mid` of the signed-in customer. The contact centre and stores look the customer up by mobile number or email and confirm identity with a one-time code sent to the registered mobile number before any change is recorded.

A contact point is a channel address: a Singapore telephone number, an email account or an app installation. Consents for the purposes MKT-SMS and MKT-VOICE are held per contact point, because the DNC provisions apply per Singapore telephone number and because a customer may have several lines (family plans average 2.3 lines per account holder). Consents for the other purposes are held per customer.

The `contact_point` table links each contact point to the `mid` that owns it. When a mobile line is terminated, the consents held on its number remain on the contact point record, which keeps the history of the number complete for audit.

Inbound withdrawals that carry only a number (a STOP reply, the IVR opt-out) are applied to the contact point for that number, so they take effect even when the caller cannot be matched to a Keruing ID.

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## 9. Consent Record Model

Izin stores consent as an append-only sequence of events per subject and purpose, where the subject is either a customer (`mid`) or a contact point. Each event has:

- `event_id` (UUID v7) and `subject_type`, `subject_id`, `purpose_code`, `purpose_version`;
- `state` (GRANTED, WITHDRAWN OR NOT\_RECORDED) and `basis`;
- `seq`, the per-subject-and-purpose sequence number, starting at 1;
- `channel`, `actor_type`, `actor_id`, `notice_version`, `evidence_ref` and `captured_at`;
- `idempotency_key`, supplied by the channel.

The Consent API writes a change in one database transaction: it reads the current `consent_state` row with `SELECT ... FOR UPDATE`, checks that the request's `expected_seq` (if supplied) matches, inserts the event with `seq + 1`, updates the state row, records the idempotency key, and inserts an outbox row. The row lock makes a lost update impossible even if two channels race: the second writer waits, then reads the new `seq`. For the first change to a subject and purpose there is no row to lock, and the primary key of `consent_state` lets only one of two concurrent first changes commit; the other is retried and then takes the lock. The idempotency key is inserted into `request_key`, whose primary key is (`actor_type`, `idempotency_key`), so a retried request returns the original result without a second event. Both keys sit on unpartitioned tables, because `consent_event` is partitioned by month (Section 18) and a unique constraint on it would have to include the partition column. Because event, state and outbox are written in one transaction, there is no dual write: a change is either fully recorded and queued for publication or not recorded at all.

`consent_state` is a projection and can be rebuilt from events by replaying them in `seq` order. A nightly job recomputes the state of a 1% random sample of subjects from their events and alerts on any difference; a full rebuild of the 8.7 million customers takes about 40 minutes on a reader instance and is run before each release that touches the projection.

A withdrawal does not delete anything: it is an event with state `WITHDRAWN`, and data use for that purpose stops. What is kept and for how long is set by Section 19, not by the state of a consent.

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## 10. Capture in Web and App

The web preference centre and the app's Privacy and Preferences screen are the same component, served by the Consent API and rendered from the purpose catalogue.

- Each consent purpose is shown on its own row, with a short statement, a link to the full notice version, and a toggle that is off until the customer turns it on. There is no "select all" control.
- Consent for any marketing or partner purpose is never a condition of signing up, buying a plan or joining Rewards; the sign-up flows complete with every toggle off.
- The notice text shown is the current notice version for the customer's language; the version ID is posted with the change, and the API rejects a preference-centre change that names a notice version which is not current for that purpose.
- Every change is sent with an idempotency key generated when the customer makes that change and reused for each retry of it, so a double tap or a network retry records one event, while a second, different change on the same screen carries a key of its own. The app or web session ID and the screen release are recorded as `evidence_ref`.
- After a change, the screen re-reads state from the API and shows the time of the change and the receipt number.

Sign-up forms in the app and on the web place the marketing toggles after the account details, under the heading "Would you like to hear from us?", with each toggle naming its channel. A customer who does nothing is recorded with no event at all for those purposes, and the Eligibility API treats an absent state as not granted.

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## 11. Capture in the Contact Centre and Stores

Contact centre agents and store staff use the Izin panel embedded in the contact-centre CRM and the point-of-sale tablet. The panel shows the same purposes as the preference centre. The agent reads the scripted notice for the purpose (the script is the notice version text), the customer answers, and the agent records the answer. The panel attaches the call recording ID (contact centre) or the signed-tablet form ID (stores) as `evidence_ref`, and the customer's one-time code confirmation from Section 8 is stored with the event.

Stores without connectivity at the time of sign-up (pop-up counters and roadshows) capture consent on the tablet offline, with the customer's signature on the tablet form as evidence in place of the one-time code; the tablet queues the changes with their original `captured_at` and idempotency keys and submits them when connectivity returns. Queued changes are submitted in the order captured, and the Consent API accepts them with the notice version that was current at their `captured_at`. If a change for the same subject and purpose has been recorded since a queued change's `captured_at`, the Consent API rejects the queued change, and the tablet tells the staff member that the customer's preference was changed elsewhere in the meantime.

Outbound telemarketing calls end with the agent offering the customer the option to stop marketing calls. A request to stop is recorded as a withdrawal of MKT-VOICE on the number called, during the call.

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## 12. Capture in Partner Channels

Registered partners call the Partner Consent API, which accepts grants and withdrawals for the partner-sharing purposes and, for Seri Assurance, also for MKT-EMAIL, because Seri's insurance sign-up journey asks customers whether they also want Keruing offers by email. The partner authenticates with mutual TLS using a client certificate issued by the Keruing partner CA, and each request is signed with the partner's private key. A partner can submit changes only for the purposes listed against it in the partner registry.

A partner identifies the customer by mobile number and date of birth. The Partner Consent API resolves these to a `mid`; if no single match is found, the change is rejected and the partner is told to direct the customer to the Keruing app. The customer confirms each change with a one-time code that Keruing sends to the registered mobile number and that the customer enters in the partner's app. Partner-captured events carry `actor_type = PARTNER` and the partner's own reference as `evidence_ref`.

Partners receive the consent state for their own partner-sharing purpose through the nightly partner file (Section 15).

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## 13. Withdrawal Semantics

A withdrawal is accepted in every channel that accepts a grant: preference centre, app, contact centre, stores, partner API, STOP reply and IVR opt-out. It is recorded in the same transaction model as a grant, and the next Eligibility API call for that subject and purpose returns "do not contact", because the Eligibility API reads consent state from the Aurora writer endpoint, not from a reader.

In the preference centre, every purpose that is on shows beside its toggle the likely consequence of turning it off (for example, "You will stop receiving Rewards double-points offers by SMS"), as section 16 of the PDPA requires. Turning the toggle off is the whole withdrawal, the same single action as turning it on: there is no confirmation step, and the withdrawal is never refused and never delayed by the consequence notice.

Withdrawal stops the use of personal data for that purpose. It does not stop service messages under SVC-NOTICE, which are not consent-based, and it does not delete the customer's data; retention is governed by Section 19. Withdrawal of MKT-SMS or MKT-VOICE on a number also ends the evidential-consent exemption for that number until the customer grants again; while the state is WITHDRAWN, the Eligibility API refuses SMS and voice marketing to the number whatever its DNC result (Section 16).

Campaigns already queued in Seruan re-check eligibility per recipient at send time (Section 15), so a withdrawal recorded minutes before a send stops that send.

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## 14. Event Bus and Change Propagation

### 14.1 Outbox

Every consent change inserts an outbox row in the same transaction as the event (Section 9). The Outbox Relay, a single active pod with a standby, reads unpublished rows in `outbox_id` order in batches of 500 and publishes them to Kafka with `acks=all` and an idempotent producer. It marks a row published only after the broker acknowledges it.

### 14.2 Topic

Consent changes are published to the topic `consent.changes` on the group's Keruing Data Hub MSK cluster (three brokers across three availability zones, replication factor 3, `min.insync.replicas` 2). The topic has 24 partitions and is keyed by subject and purpose (`subject_id:purpose_code`), with `cleanup.policy=compact`, `segment.ms` of one hour and `min.compaction.lag.ms` of one hour, so the topic keeps the latest event for every subject and purpose indefinitely and a new consumer can bootstrap from it without a database extract. For a contact point, the subject is the contact point ID.

The Keruing Data Hub cluster is shared by the group's data products. Any service holding a client certificate issued by the Data Hub certificate authority can consume any topic on it, which keeps onboarding of new consumers to a single certificate request.

### 14.3 Event payload

Each event describes a single change: the subject, one purpose, the new state, the `seq`, the channel and actor, and the effective time. Each event also carries the customer's full profile snapshot (name, NRIC number, date of birth, mobile number, email and postal address), so that consumers never need to call back into Izin to act on an event.

Events use a versioned Avro schema in the Data Hub schema registry with backward-compatible evolution only.

### 14.4 Relay failure handling

The relay attempts to publish each outbox row up to five times with exponential backoff from 200 ms. A row that still fails is marked `FAILED` and the relay moves on to the next row, so one malformed or oversized event never blocks the stream. `FAILED` rows are listed on the operations dashboard and replayed after review at the weekly operations meeting.

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## 15. Downstream Consumers

| Consumer                 | Owner               | How it consumes                                                                     | What it does with a change                                                                                    |
|--------------------------|---------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Seruan campaign platform | Group Marketing     | Real-time consumer group; also calls the Eligibility API per recipient at send time | Updates audience membership; the send-time check is the final gate                                            |
| Group Analytics Platform | Group Data Office   | Batch consumer, every 6 hours                                                       | Excludes withdrawn customers from PRS-OFFERS models and segment exports                                       |
| Contact centre CRM       | Customer Operations | Real-time consumer group                                                            | Shows current consent state on the agent desktop; outbound dial lists call the Eligibility API                |
| Rewards platform         | Keruing Rewards     | Real-time consumer group                                                            | Updates newsletter and push audiences                                                                         |
| Partner Gateway          | Partnerships        | Batch consumer, nightly at 01:00 SGT                                                | Builds the nightly partner files of grants and withdrawals for PTN-INS and PTN-CARD and delivers them by SFTP |

Consumers apply events in the order received. Seruan and the contact centre CRM treat any non-200 response or timeout from the Eligibility API as "do not contact" (P5), and Seruan records the skipped recipients with the reason so that Marketing can see the effect.

Each partner file contains, per customer whose partner-sharing state changed since the previous file, the name, mobile number, email, Rewards tier and the new state. The partner contract requires the partner to stop using the customer's details for marketing within one business day of a withdrawal appearing in a file.

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## 16. Do Not Call Registry Checks

The DNC Service keeps, for every Singapore telephone number in the marketing base, the most recent DNC Registry result for the voice, text message and fax registers. It submits numbers to the PDPC's DNC Registry through the bulk-check interface and stores each result with the date it was received.

Under the PDPA, the result of a DNC Registry check remains valid for 21 days from receipt, so the DNC Service re-checks every number in the marketing base once every 14 days and serves campaign checks from its cache. The base of about 7.4 million numbers is split into 14 daily cohorts by a hash of the number, so each night's bulk submission is about 530,000 numbers and completes within the overnight window. A number whose result is older than 21 days because its cohort's refresh has not succeeded keeps its last result and is served from it until the next successful refresh, so a registry outage or a rejected bulk file never blocks a campaign. New numbers added to the marketing base are checked within the same day through the single-number interface before they are eligible for any marketing SMS or call.

The Eligibility API combines consent and DNC status for SMS and voice as follows:

| Consent state for MKT-SMS / MKT-VOICE on the number             | DNC result                                                 | Decision                                                   |
|-----------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|
| GRANTED, with an evidence_ref                                   | Any                                                        | Allowed (clear and unambiguous consent in evidential form) |
| GRANTED, without an evidence_ref (migrated records, Section 25) | Not registered on the relevant register in the result held | Allowed                                                    |
| GRANTED, without an evidence_ref                                | Registered, or no result held                              | Not allowed                                                |
| WITHDRAWN or no state                                           | Any                                                        | Not allowed                                                |

A GRANTED state captured through Izin always carries an evidence reference (Sections 10 to 12) and is clear and unambiguous consent in evidential form, so marketing to the number is allowed even if it is listed on the DNC Register. For GRANTED states without one, which come only from the migration, the DNC result decides. Izin never allows marketing on DNC status alone. Email and app push are not covered by the DNC provisions and depend on consent only.

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## 17. Consent Receipts

A consent receipt is a record, given to the customer, of a consent change: what changed, when, through which channel, and the notice version that applied. The Receipt Service creates a receipt for each change request (a request that changes several purposes at once gets one receipt listing all of them).

Each receipt has a receipt number of the form R followed by a ten-digit sequence (for example R0004417302), allocated from a Postgres sequence, for reference in calls and letters. The receipt link carries a separate random 128-bit token (<https://izin.keruing.sg/r/{token}>), stored only as its SHA-256 hash, and the link expires 30 days after issue. Without signing in, the receipt page shows the customer's first name, the masked mobile number or email the receipt was sent to, the receipt number, and the purposes changed with their new states and the time of the change. The current state of every other purpose, and the full contact details, are shown only after the customer signs in with Keruing ID.

Receipts are sent by SMS to the mobile number on the profile, or by email where the customer has no mobile number. Receipts are not themselves marketing and are sent under SVC-NOTICE.

---

## 18. Data Model

```
CREATE TABLE purpose (  
    purpose_code    text PRIMARY KEY,
```

```

basis            text NOT NULL,          -- CONSENT, CONTRACT, LEGAL

subject_type     text NOT NULL,          -- CUSTOMER or CONTACT_POINT

current_version  int  NOT NULL,

retired_at       timestamptz

);

CREATE TABLE notice_version (

  purpose_code    text NOT NULL REFERENCES purpose,

  purpose_version int  NOT NULL,

  notice_version  text NOT NULL,

  material        boolean NOT NULL,

  effective_from  timestamptz NOT NULL,

  text_en text NOT NULL, text_zh text NOT NULL, text_ms text NOT NULL, text_ta text NOT NULL,

  PRIMARY KEY (purpose_code, notice_version)

);

CREATE TABLE contact_point (

  contact_point_id uuid PRIMARY KEY,

  kind              text NOT NULL,          -- MSISDN, EMAIL, APP_INSTALL

  address           text NOT NULL,          -- encrypted column (Section 21)

  address_hash      bytea NOT NULL UNIQUE,  -- HMAC-SHA256 lookup key

  mid              uuid,

  linked_at         timestamptz

);

CREATE TABLE consent_event (

  event_id         uuid NOT NULL,

  subject_type     text NOT NULL,

  subject_id       uuid NOT NULL,

  purpose_code     text NOT NULL REFERENCES purpose,

  purpose_version  int  NOT NULL,

  seq              int  NOT NULL,

  state            text NOT NULL,

```

```

channel          text NOT NULL,
actor_type       text NOT NULL,
actor_id         text NOT NULL,
notice_version   text NOT NULL,
evidence_ref     text,
idempotency_key  text NOT NULL,
captured_at      timestampz NOT NULL,
stored_at        timestampz NOT NULL DEFAULT now(),
PRIMARY KEY (event_id, stored_at)
) PARTITION BY RANGE (stored_at);

```

```

CREATE TABLE consent_state (
    subject_type    text NOT NULL,
    subject_id      uuid NOT NULL,
    purpose_code    text NOT NULL,
    purpose_version int  NOT NULL,
    state           text NOT NULL,
    seq             int  NOT NULL,
    last_event_id   uuid NOT NULL,
    updated_at      timestampz NOT NULL,
    PRIMARY KEY (subject_type, subject_id, purpose_code)
);

```

```

CREATE TABLE request_key (
    actor_type      text NOT NULL,
    idempotency_key text NOT NULL,
    event_ids       uuid[] NOT NULL,
    created_at      timestampz NOT NULL DEFAULT now(),
    PRIMARY KEY (actor_type, idempotency_key)
);

```

```

CREATE TABLE receipt (
    receipt_number text PRIMARY KEY,

```

```

mid          uuid,

event_ids    uuid[] NOT NULL,

sent_to      text NOT NULL,

created_at   timestampz NOT NULL

);

CREATE TABLE outbox (

  outbox_id   bigserial PRIMARY KEY,

  event_id    uuid NOT NULL,

  payload     bytea NOT NULL,

  status      text NOT NULL DEFAULT 'PENDING', -- PENDING, PUBLISHED, FAILED

  attempts    int  NOT NULL DEFAULT 0,

  created_at  timestampz NOT NULL DEFAULT now()

);

```

consent\_event is partitioned by month of stored\_at. Customer profile fields used in events and receipts (name, NRIC number, date of birth, postal address) are read from the group customer master at write time and are not stored in Izin tables.

## 19. Retention and Archival

| Data                    | Retention                                | Mechanism                                                  |
|-------------------------|------------------------------------------|------------------------------------------------------------|
| consent_state           | While the subject exists, then 12 months | Deleted 12 months after the customer's last account closes |
| consent_event           | 24 months                                | Monthly partition drop                                     |
| receipt                 | 24 months                                | Monthly purge job                                          |
| outbox (published rows) | 7 days                                   | Daily purge job                                            |
| request_key             | 90 days                                  | Daily purge job                                            |
| DNC results             | 90 days after the result's receipt       | Daily purge job                                            |
| Database backups        | 35 days                                  | Aurora automated backups                                   |

Consent events older than 24 months are deleted by dropping their monthly partition, and the current position for every subject and purpose remains in consent\_state, so Izin always knows what each customer has agreed to. Dropping a partition is instant and avoids the vacuum load of row-by-row deletes on the largest table.

Contact-centre call recordings are retained by the contact-centre platform under its own policy; signed store forms are retained by the stores document archive.

## 20. Regulator Audit and Evidence Export

When the Commission investigates a complaint, or a customer disputes a marketing contact, the ODPO must show that Keruing had consent (or a valid DNC result) at the time of the contact. The evidence export produces, for the subjects, purposes and time range requested:

- the consent history: every event with its channel, actor, notice version, evidence reference and time;
- the exact notice text of every notice version referenced;

- the DNC results held for the numbers concerned, with their dates of receipt;
- the receipts issued.

The export is a signed ZIP (detached signature with the ODPO export key in AWS KMS) containing CSV files and a manifest with SHA-256 hashes, generated by an asynchronous job from the ODPO Console. Requests are logged with the requester and the reason. For a single customer the export completes in under a minute; for a whole purpose it runs overnight.

The ODPO's internal target is to answer a Commission request within five business days (NFR-5).

---

## 21. Security and Access Control

- **Authentication of callers.** First-party channels call the Consent API with OAuth 2.0 client credentials issued by the group identity provider, plus the end customer's access token for customer-initiated changes. Agent and staff changes carry the agent's identity from the CRM or POS session. Partners use mutual TLS and signed requests (Section 12).
  - **Authorisation.** Each channel client is allowed only the actor types it may claim; the web client cannot submit a change as AGENT, and the contact-centre client cannot submit as CUSTOMER.
  - **Encryption.** TLS 1.2 or higher in transit. Aurora storage encryption with a customer managed KMS key. Contact point addresses are additionally encrypted at column level with a data key from KMS, and looked up by an HMAC of the normalised address.
  - **Staff access.** ODPO Console users sign in through corporate SSO with phishing-resistant MFA. Every view of a customer record and every export is written to the access log, which is shipped to the group security log store and retained for seven years.
  - **Database access.** No human has standing access to the production database; break-glass access requires a second approver and is time-limited to four hours.
- 

## 22. Data Residency and Partner Data Sharing

NFR-7 places every Izin data store in Singapore: the Aurora cluster, the MSK cluster, backups, logs and the evidence export bucket are all in ap-southeast-1, and Izin has no other region.

Seri Assurance receives its nightly partner file on its SFTP server in Kuala Lumpur, where its marketing and contact-centre systems run. Lintang Bank receives its file on an SFTP server in its Singapore data centre. Residency is fully met by this placement: every Izin data store is in ap-southeast-1, and partner processing after the nightly file is delivered is the partner's own responsibility.

Partners are registered in the ODPO Console with their purposes, SFTP endpoints, PGP keys and contact persons. A partner can be suspended in the console, which stops its file and rejects its Partner Consent API calls at once.

---

## 23. Resilience and Disaster Recovery

### 23.1 Within the region

The Consent Store is an Aurora PostgreSQL cluster with a writer and two readers in three availability zones. Aurora writes six copies of each change across three AZs and acknowledges a commit once four are durable, so the loss of one AZ loses no acknowledged change (NFR-4); failover to a reader takes about 30 seconds, during which the Consent API returns 503 and channels retry with the same idempotency key. The MSK cluster has three brokers across three AZs, replication factor 3, `min.insync.replicas 2` and `producer acks=all`, so a published event survives the loss of an AZ. EKS runs every Izin API service with at least three replicas spread across AZs, and the Outbox Relay's active and standby pods in different AZs.

### 23.2 Region loss

Izin has no second region, by decision DEC-04: AWS has one region in Singapore, and NFR-7 rules out placing a

replica elsewhere. Aurora snapshots are copied every four hours to a backup vault in a separate AWS account in ap-southeast-1 with vault lock, so that a compromise of the production account cannot destroy them. In the event of a full loss of ap-southeast-1, Izin is rebuilt from the latest snapshot when the region returns; the business has accepted an RTO of 24 hours and an RPO of four hours for that event, and recorded it in the risk register (RR-118).

### 23.3 Behaviour during an outage

While the Consent API is unavailable, channels show the customer that preferences cannot be changed right now and do not queue changes, except the offline store tablets (Section 11). While the Eligibility API is unavailable, Seruan and the contact centre stop marketing contacts (P5); service messages, which do not depend on Izin, continue.

## 24. Latency and Capacity Budget

### 24.1 Consent API latency (p99, single change)

| Step                                                                                                     | Budget                        |
|----------------------------------------------------------------------------------------------------------|-------------------------------|
| API gateway, TLS termination and token validation                                                        | 25 ms                         |
| Request validation against the cached purpose catalogue                                                  | 5 ms                          |
| Database transaction (lock state row, insert event, update state, record idempotency key, insert outbox) | 60 ms                         |
| Customer master read for the profile snapshot                                                            | 40 ms                         |
| Receipt number allocation and receipt row                                                                | 15 ms                         |
| Response                                                                                                 | 5 ms                          |
| <b>Total</b>                                                                                             | <b>150 ms</b> (NFR-1: 300 ms) |

### 24.2 Eligibility API capacity

The heaviest load on Izin comes from campaign sends, because Seruan checks every recipient with the Eligibility API at send time. The largest campaign, the monthly Rewards newsletter, goes to 3.0 million recipients in a one-hour send window, which is about 833 requests per second. The Eligibility API is provisioned for 1,000 requests per second at p99 50 ms (NFR-8), and Seruan paces each campaign to at most 800 requests per second, so a newsletter of 3.0 million recipients completes in about 63 minutes; requests above the provisioned rate are rejected with 429 by the API gateway's rate limit to protect the Consent Store writer.

### 24.3 Storage

| Table         | Basis                                                  | Size         |
|---------------|--------------------------------------------------------|--------------|
| consent_state | 5.2 million customers × 7 consent purposes × 180 bytes | about 6.6 GB |
| consent_event | 140,000 events a day × 24 months × 600 bytes           | about 61 GB  |
| contact_point | 12 million contact points × 300 bytes                  | about 3.6 GB |
| receipt       | 140,000 receipts a day × 24 months × 450 bytes         | about 46 GB  |

The Aurora cluster uses db.r7g.2xlarge instances; the working set of consent\_state and the indexes fits in memory.

## 25. Migration from Legacy Systems

The legacy sources are the telco CRM (Arus), the Rewards platform (Mata) and the scanned store sign-up forms. Migration runs once per business unit, before that unit's channels switch to Izin.

| Legacy source | Legacy field                                               | Izin mapping                               |
|---------------|------------------------------------------------------------|--------------------------------------------|
| Arus          | sms_optout = Y                                             | MKT-SMS WITHDRAWN on the line's number     |
| Arus          | call_optout = Y                                            | MKT-VOICE WITHDRAWN on the line's number   |
| Arus          | email_optout = Y                                           | MKT-EMAIL WITHDRAWN                        |
| Arus          | sms_optin = Y or call_optin = Y (roadshow sign-ups 2016 to | MKT-SMS or MKT-VOICE GRANTED on the line's |

| Legacy source | Legacy field                                  | Izin mapping                                               |
|---------------|-----------------------------------------------|------------------------------------------------------------|
|               | 2019; paper forms not retained)               | number, no evidence_ref                                    |
| Mata          | news letter = Y with sign-up date and channel | MKT-EMAIL GRANTED, evidence_ref to the Mata sign-up record |
| Store forms   | Ticked marketing box, signed                  | Purposes ticked GRANTED, evidence_ref to the form scan     |

Keruing Mobile's 2019 privacy notice told all subscribers that Keruing would market its products by SMS, call, email and push unless they opted out. Customers with no recorded preference in the legacy systems are migrated with no event for the marketing purposes, so the Eligibility API treats those purposes as not granted (Section 10). They are invited to set their preferences through a card in the Keruing app and a notice with their monthly bill, both of which link to the preference centre. The marketing base reachable on the first day is therefore about 1.9 million customers, not the 3.4 million reachable from the legacy flags; Group Marketing has accepted this.

Every migrated event has channel = MIGRATION, actor\_type = SYSTEM and the captured\_at of the legacy record where one exists, else the migration time. The migration is idempotent: it uses the legacy record key as the idempotency key, so it can be re-run after a partial failure.

## 26. Confirmed Decisions

| ID     | Decision                                                                                                                    | Rationale                                                             |
|--------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| DEC-01 | Aurora PostgreSQL 16, db.r7g.2xlarge, writer and two readers, three AZs                                                     | Transactional writes of event, state and outbox; familiar to the team |
| DEC-02 | Amazon MSK (Keruing Data Hub cluster), topic consent.changes, 24 partitions, compacted, keyed by subject and purpose        | Shared platform, bootstrap from the topic                             |
| DEC-03 | Transactional outbox with a single active relay                                                                             | No dual write                                                         |
| DEC-04 | Single region ap-southeast-1; snapshot copies to a separate account; region-loss RTO 24 h, RPO 4 h accepted (RR-118)        | NFR-7; only one AWS region in Singapore                               |
| DEC-05 | MKT-SMS and MKT-VOICE held per contact point; other purposes per customer                                                   | DNC provisions apply per number                                       |
| DEC-06 | DNC results cached and refreshed in 14 daily cohorts; a result awaiting refresh is served until the next successful refresh | 21-day validity; registry load                                        |
| DEC-07 | Eligibility API reads consent state from the writer endpoint                                                                | Read-your-writes for withdrawals                                      |
| DEC-08 | Receipts by SMS, or email where no mobile number exists                                                                     | Every customer has at least one                                       |
| DEC-09 | Seri Assurance captures Keruing consent (PTN-INS and MKT-EMAIL) in its own app through the Partner Consent API from Phase 4 | Partner programme launch in Q2 2027                                   |
| DEC-10 | Partner files nightly by SFTP with PGP encryption                                                                           | Partner capabilities                                                  |
| DEC-11 | Legacy preferences migrated per Section 25; no consent is created for customers with no recorded preference                 | Only recorded choices carry over                                      |
| DEC-12 | Consent events retained 24 months                                                                                           | Storage and vacuum load                                               |

## 27. Pending Backlog

Items confirmed as in scope but not yet fully designed:

| ID    | Item                                                                                                                                                               | Status                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| PB-01 | Chinese, Malay and Tamil notice translations for PTN-INS and PTN-CARD                                                                                              | With the translation vendor                         |
| PB-02 | Seruan dashboard of recipients skipped by the Eligibility API, by reason                                                                                           | Seruan team scheduled for Q1 2027                   |
| PB-03 | Data-sharing agreement with Seri Assurance, including the legal opinion on whether consent captured by Seri for Keruing purposes is valid consent given to Keruing | Pending with Group Legal; opinion requested 2026-08 |
| PB-04 | Customer-facing receipt page design and accessibility review                                                                                                       | Design team                                         |
| PB-05 | Retirement plan for the Arus and Mata opt-out columns after migration                                                                                              | Business units                                      |

| ID    | Item                                            | Status |
|-------|-------------------------------------------------|--------|
| PB-06 | ODPO Console role model beyond Admin and Viewer | ODPO   |

## 28. Validation and Acceptance Criteria

Each requirement from Section 2 is validated by a specific method with a concrete pass or fail criterion. This table is the basis for the test plan in Build Phase 6.

### 28.1 Functional requirements

| ID    | Validation method                           | Acceptance criteria                                                                                                                                                                                                                                                                                        |
|-------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1  | API contract tests per channel client       | Every channel can grant and withdraw every purpose it is registered for; requests for other purposes return 403.                                                                                                                                                                                           |
| FR-2  | UI review and API negative tests            | No sign-up or preference screen has a pre-selected marketing toggle; the API rejects a grant without a current notice version.                                                                                                                                                                             |
| FR-3  | Event completeness check                    | For 1,000 changes across all channels, every event has channel, actor, notice version, time and evidence reference populated.                                                                                                                                                                              |
| FR-4  | Channel walkthrough                         | In every channel, withdrawing a purpose takes no more steps than granting it.                                                                                                                                                                                                                              |
| FR-5  | Outbox and relay test                       | Every committed change appears on consent . changes exactly once (by event_id) after a relay pod is killed mid-batch.                                                                                                                                                                                      |
| FR-6  | Eligibility decision table test             | For every combination in Section 16, the API returns the documented decision.                                                                                                                                                                                                                              |
| FR-7  | DNC test with the registry test environment | Numbers registered in the test registry are refused without evidential consent and allowed with it.                                                                                                                                                                                                        |
| FR-8  | Preference centre end-to-end test           | Every purpose is listed with its state and notice text; a change made in the app is shown on the web after reload.                                                                                                                                                                                         |
| FR-9  | Receipt test                                | For 500 changes across every channel, each produces exactly one receipt within five minutes, sent to the mobile number or, for profiles with no mobile number, the email; the receipt link opens the receipt without sign-in, shows only the fields listed in Section 17, and stops working after 30 days. |
| FR-10 | Partner API conformance test                | A registered partner can submit only its listed purposes; an unregistered certificate is refused.                                                                                                                                                                                                          |
| FR-11 | Evidence export test                        | For 50 seeded customers, the export contains every event, the notice texts and the DNC results, and the signature verifies.                                                                                                                                                                                |
| FR-12 | Keyword and IVR test                        | A STOP reply and the IVR opt-out each produce a withdrawal on the number within one minute.                                                                                                                                                                                                                |
| FR-13 | Migration reconciliation                    | Counts of migrated grants and withdrawals per source match the legacy extracts.                                                                                                                                                                                                                            |
| FR-14 | Console test                                | ODPO users can create a purpose version and a notice version; viewers cannot.                                                                                                                                                                                                                              |

### 28.2 Non-functional requirements

| ID    | Validation method                                        | Acceptance criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1 | Load test at 120 changes per second                      | p99 Consent API latency of 300 ms or less over 30 minutes.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| NFR-2 | Propagation test in pre-production at design-target load | Over 24 hours, record 1,000 withdrawals spread across every channel (app, web, contact centre, store, partner API, STOP reply and IVR), a third of them followed within an hour by another change for the same customer; restart the relay and each consumer once during the run. Every withdrawal is applied in Seruan, the Group Analytics Platform, the contact centre CRM and the Rewards platform within 24 hours of being recorded, and every partner-purpose withdrawal appears in the next nightly partner file. |
| NFR-3 | Synthetic probes                                         | Monthly availability of 99.95% or more measured by probes every 30 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| NFR-4 | AZ failure game day                                      | After a forced writer failover under load, every change acknowledged to the load generator is present in consent_event.                                                                                                                                                                                                                                                                                                                                                                                                  |
| NFR-5 | Evidence drill                                           | The ODPO produces a full evidence export for 10 named customers within five business days.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| NFR-6 | Security review and penetration                          | No high or critical findings open at go-live; access log entries present for every console view in the                                                                                                                                                                                                                                                                                                                                                                                                                   |

| ID    | Validation method                                                                                                      | Acceptance criteria                                                                                                        |
|-------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|       | test                                                                                                                   | test.                                                                                                                      |
| NFR-7 | Infrastructure review                                                                                                  | Every Izin resource in the IaC is in ap-southeast-1.                                                                       |
| NFR-8 | Load test of the Eligibility API at 1,000 requests per second, with a concurrent consent-change load of 120 per second | p99 latency of 50 ms or less over 30 minutes, with no 429 responses below 1,000 requests per second.                       |
| NFR-9 | DNC compliance audit                                                                                                   | For a sample of 1,000 marketing SMS from the pilot, each number had evidential consent or a valid DNC result at send time. |

## 29. Implementation Readiness Assessment

Could an engineering team build each component from this document without further design decisions?

| Component                        | Ready?       | What remains to decide                           |
|----------------------------------|--------------|--------------------------------------------------|
| Consent API and record model     | Ready        | Sections 9, 18.                                  |
| Preference centre (web and app)  | Ready        | Section 10; receipt page design in PB-04.        |
| Contact centre and store capture | Ready        | Section 11.                                      |
| Partner Consent API              | Ready        | Section 12; Seri go-live per DEC-09.             |
| Outbox Relay and event bus       | Ready        | Section 14.                                      |
| Eligibility API and DNC Service  | Ready        | Section 16.                                      |
| Receipt Service                  | Mostly ready | Page design (PB-04).                             |
| Partner Gateway                  | Ready        | Section 15.                                      |
| Retention jobs                   | Ready        | Section 19.                                      |
| Evidence export                  | Ready        | Section 20.                                      |
| Migration                        | Ready        | Section 25.                                      |
| Seruan integration               | Mostly ready | Section 15; skipped-recipient dashboard (PB-02). |

### Overall conclusion

The consent core, propagation and DNC handling are specified to implementation level. The open items (translations, the skipped-recipient dashboard, the receipt page design and the console role model) do not block the first cohort.

## 30. Build Phases

| # | Phase                                                                                                                                               | Depends on open design items?                  |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 1 | Foundations: EKS namespace, Aurora cluster, topic and schema registry, IaC, CI/CD, observability                                                    | No, ready                                      |
| 2 | Consent core: Consent API, record model, outbox and relay, preference centre for Keruing Mobile, receipts                                           | Partially, receipt page design (PB-04)         |
| 3 | Eligibility API, DNC Service, Seruan and contact centre CRM integration; migration of Arus                                                          | Partially, skipped-recipient dashboard (PB-02) |
| 4 | Stores, Rewards and Fibre channels; migration of Mata and store forms; Partner Consent API and Partner Gateway with Seri Assurance and Lintang Bank | No, ready                                      |
| 5 | Evidence export, ODPO Console, retention jobs                                                                                                       | No, ready                                      |
| 6 | Hardening: load, failure and DR game days; penetration test; Section 28 test plan                                                                   | Follows Section 28                             |

The first cohort (Keruing Mobile in web and app) goes live after Phase 3; the other business units follow at six-week intervals. No implementation has begun.